BTS–VI(R/S)–05–24–2431 Reg. No. 2 0 5 ...

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BTS–VI(R/S)–05–24–2443 Reg. No.

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0602 MACHINE DESIGN I

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the customers' need, formulate the problem and draw the design specifications.

CO2: Understand component behaviour subjected to loads and identify the failure criteria.

CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.

CO4: Analyse the stresses and strains induced in a machine element.

CO5: Design machine components using theories of failure.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		Question	Marks	BL	CO	PO
		$(8 \times 3 = 24)$				
I.	(a)	What are the factors to be considered for the selection of materials for the design of machine elements? Discuss.	3	L1	2	1
	(b)	What are fits and tolerances? How are they designated?	3	L1	1	1
	(c)	Define the following terms : (i) Major diameter (ii) Minor diameter (iii) Pitch (iv) Lead.	3	L2	1	1
	(d)	Distinguish between cotter joint and knuckle joint.	3	L2	1	1
	(e)	What is the difference between caulking and fullering? Explain with the help of neat sketches.	3	L2	2	1
	(f)	Discuss the materials and practical applications for the various types of springs.	3	L2	1	1
	(g)	What do you understand by the term welded joint? How it differs from riveted joint?	3	L2	2	1
	(h)	What do you understand by torsional rigidity and lateral rigidity?	3	L1	2	1

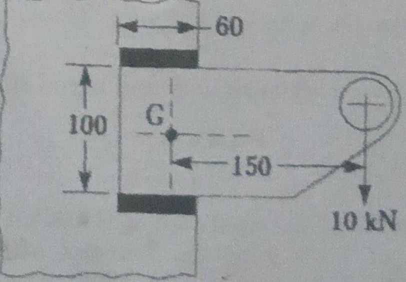
PART B

	Question	Marks	BL	CO	PO
	$(4 \times 12 = 48)$				
II.	The load on a bolt consists of an axial pull of 10 kN together with a transverses hear force of 5 kN. Find the diameter of bolt required according to: (i) Maximum principal stress theory (ii) Maximum shear stress theory (iii) Maximum principal strain theory (iv) Maximum strain energy theory (v) Maximum distortion energy theory.	12	L4	5	1,2,3

OR

(P.T.O.)

			Marks	BL	CO	PO
III.	(a)	A hollow shaft of 40 mm outer diameter and 25 mm inner diameter is subjected to a twisting moment of 120 N-m, simultaneously; it is subjected to an axial thrust of 10 kN and a bending moment of 80 N-m. Calculate the maximum compressive and shear stresses.	6	L3	3	1,2,3
	(b)	Calculate the tolerances, fundamental deviations and limits of sizes for the shaft designated as 40 H8 / f7.	6	L3	3	1,2,3
IV.		Design and draw a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa Shear stress = 35 MPa and Crushing stress= 90 MPa	12	L4	4	1,2,3
		OR				
V.		Design a knuckle joint to connect two mild steel bars under a tensile load of 25 kN. The allowable stresses are 65 MPa in tension, 50 MPa in shear and 83 MPa in crushing.	12	L4	4	1,2,3
VI.		Design a double riveted double strap butt joint for the longitudinal seam of a boiler shell, 750 mm in diameter, to carry a maximum steam pressure of 1.05 MPa gauge. The allowable stresses are: $\sigma_t = 35$ MPa $\tau = 28$ MPa $\sigma_c = 52.5$ MPa Assume the efficiency of the joint as 75%.	12	L4	3	1,2,3
		OR				
VII.		A helical valve spring is to be designed for an operating load range of approximately 90 to 135 N. The deflection of the spring for the load range is 7.5 mm. Assume a spring index of 10. Permissible shear stress for the material of the spring = 480 MPa and its modulus of rigidity = 80 GPa. Design the spring.	12	L4	4	1,2,3
VIII.		A bracket, as shown in Figure, carries a load of 10 kN. Find the size of the weld if the allowable shear stress is not to exceed 80 MPa.	12	L4	3	1,2,3



All dimensions in mm.

OR

			Marks	BL	CO	PO
IX.	(a)	A hollow shaft for a rotary compressor is to be designed to transmit a maximum torque of 4750 N-m. The shear stress in the shaft is limited to 50 MPa. Determine the inside and outside diameters of the shaft, if the ratio of the inside to the outside diameter is 0.4.	6	L3	4	1,2,3
	(b)	Under what circumstances are hollow shafts preferred over solid shafts? Give any two examples where hollow shafts are used. How they are generally manufactured?	6	L2	2	1

Blooms's Taxonomy Level's
L1 – 7.50%, L2 – 17.5%, L3 – 15%, L4 – 60%.

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